Resonance

Three buildings, three natural frequencies — and a game you cannot win by shaking harder

Level: Scientist — year XI (ages 16–18) • **Time in class:** 20 min as a hook, or a 50-minute lesson • **Sensors:** MPU-6500 (acceleration)



The activity screen. The values shown are simulated; yours will differ.

Before you start

Who it is for

Scientist — year XI (ages 16–18). In **year XI**, mechanical oscillations and resonance are explicit physics syllabus content, and the activity becomes a quantitative demonstration. The idea of a period comes from lower secondary (periodic motion); as an opening game it can be shown earlier, but the app lists it at the Scientist level only. The curriculum plan classes it as an *engineering extension* of the seismic zone.

Assumed knowledge

The idea of a **period** and a **frequency**, and the link between them ($T = 1/f$). For the advanced discussion: damped oscillation and the notion of a quality factor.

Time in class

Short: 20 min as a hook. Full: a 50-minute lesson, with the periods measured, the *harder does not help* test, and the link to earthquake engineering.

Teacher preparation

5 min — play one round and **win all three stars yourself**, so that you know how different the rhythms have to be; and have a metronome or a metronome app ready if you want the quantitative version.

Sensor settling time

None. The accelerometer responds instantly. **But the buildings need a few seconds** to build up their swing — resonance grows over time rather than appearing at once, and that is itself an observation worth making.

Sensors and boxes

MPU-6500. One box is enough, with the screen projected and the children taking turns. The box is held **flat** and moved *sideways* — it is not tilted.

Three routes

Short	20 min: the prediction, two or three rounds trying for all three stars, and the debrief — why shaking harder does not help.
Full	The 50-minute lesson: the short route, plus **measuring the periods** with a stopwatch or a metronome, checking the $T \approx N/10$ rule, and the link to earthquake-resistant design.
Project	Extension: each pupil works out the frequency their own block of flats would have and says which shaking rhythm would be most dangerous for it.

1. Where this fits the curriculum

- **Physics** — in lower secondary, periodic motion, period and frequency (the exact year depends on the syllabus — check the current one): the three buildings have their periods printed on screen (1.0 s, 0.5 s and 0.3 s), and a child can *beat them out by hand*. In **year XI**, mechanical oscillations, damped and forced oscillation and **resonance** are explicit content, and here they have a demonstration anyone can perform.
- **Mathematics** — the relation $T = 1/f$, used in both directions; and inverse proportionality in a case where you can *see* it: the taller the building, the lower its frequency.
- **Design and technology** — structural design: why buildings of the same height behave alike in an earthquake, and what engineers do to change a structure's natural frequency.
- **Geography** — in **year V** and **year IX**, earthquakes. **In Romania the link is direct and painful**: in the 1977 earthquake, buildings of a particular height suffered disproportionately, because their own period matched that of the ground motion.
- **Across subjects** — the idea of *tuning*: a system responds strongly only at one frequency. The same idea explains a radio, a glass shattered by a voice, and a swing pushed at the right moment.

The central skill is **understanding selectivity**. The game is not won by shaking harder but by shaking *at the right rhythm* — and the pupil who can say why has understood resonance better than any definition would teach them.

2. Timing

Route	Time	What happens
Hook	**20 min**	The prediction (3), one projected round in which you try it yourself (3), the pupils' rounds, taking turns (8), the *harder does not help* test (3), the debrief (3).
Full	**50 min**	The short route (20), measuring the periods by counting or with a metronome (10), checking the $T \approx N/10$ rule against real buildings in town (10), the link to earthquake-resistant design and the conclusion (10).

The star is not won by force — and that is the whole lesson

The rule of the game is subtler than it looks, and is worth knowing in advance: **a star is won when one building dances *and* clearly outstrips the other two**, for over a second. It is not enough for it to move a lot; the others must move *little*.
The consequence is that **hard, chaotic shaking wins no stars at all**: it excites all three buildings at once, none pulls ahead, and the screen stays starless however violently you rattle it.

The only strategy that works is a clean rhythm: a slow, regular motion — about a second there and back — wakes the tower; one twice as fast wakes the block; one more than three times as fast, the house. That is why the prompt on screen is **"Make each building dance on its own"**: one at a time, not all together. The star rule is now also written on screen, just under the header, but children do not read the small line before they start shaking. Tell them it is **a game of rhythm, not of strength**, but tell them *after* the first round, reading the line under the header together — finding it out for themselves is far more powerful.

3. Materials

What you need	Per group	Notes
MagicBox with MPU-6500	1	Held **flat** and moved sideways, not tilted.
Laptop with the app	1	**WiFi is preferable**: the motion is broad and repeated, and the cable gets in the way.
Metronome, or a metronome app	1	For the quantitative version: 60 beats per minute = 1 Hz, which is exactly the tower. **It turns the game into a measurement.**
Stopwatch	1	For counting periods by hand.
String and three weights	1 set (optional)	**Three pendulums of different lengths** are the best physical companion: tie them to the same ruler, move the ruler sideways, and only the one whose length matches the rhythm dances. It is the same phenomenon without a screen.
Neck strap or wrist strap	1	The box moves a great deal; a strap prevents a fall.

Preparation. Play one round and win all three stars — you will find out for yourself how *different* the three rhythms have to be, and you will be able to tell the class with authority that it can be done. If you are doing the quantitative version, set the metronome to 60, 120 and 200 beats per minute — that is 1, 2 and 3.3 Hz.

4. Science background for the teacher

The forced oscillator, and why it answers only one rhythm

Each building in the game is simulated as a **damped harmonic oscillator**, driven through its base by the motion of your hand:

$$\ddot{x} + 2\zeta\omega_0\dot{x} + \omega_0^2 x = -a_{\text{ground}}(t),$$

where $\omega_0 = 2\pi f_0$ is the natural angular frequency and ζ the damping ratio.
When the drive is harmonic, at frequency f, the amplitude of the response has a sharp maximum at $f \approx f_0$. At resonance, the amplification relative to the base motion is given by the **quality factor**

$$Q = \frac{1}{2\zeta}.$$

The app uses $\zeta = 0.05$, so $Q = 10$: at the right rhythm, the top of the building moves about ten times as much as your hand. That is why a small rhythmic motion lifts the tower and a large chaotic one lifts nothing.

The low damping is not an arbitrary choice: it is what makes the resonance **sharp**, that is, selective. A real building is damped rather more — typically ζ between 0.02 and 0.05 for concrete structures in the elastic range — so the effect is realistic.

4.1 Why tall buildings sway more slowly

The three buildings in the game have periods of 1.0 s, 0.5 s and 0.3 s, for 10, 5 and 3 storeys. These are not invented figures: they follow exactly a rule of thumb used by engineers,

$$T \approx \frac{N}{10} \text{ seconds},$$

where N is the number of storeys. A ten-storey block sways with a period of about a second; a three-storey house, about three tenths of a second.

The physical reason is that the natural period grows with mass and falls with stiffness,

$$T = 2\pi\sqrt{\frac{m}{k}},$$

and in a taller building the mass grows faster than the stiffness. The $N/10$ rule is approximate, but good enough for pupils to check it against buildings in their own town by counting the storeys.

This is the activity's best piece of homework: count the storeys of the block you live in, estimate its period, and say which shaking rhythm would be most dangerous for it.

4.2 The link with earthquakes in Romania

Ground motion in an earthquake does not have a single frequency but a spectrum — and that spectrum depends on the source, on the distance and, very much, **on the ground the building stands on**. A thick layer of soft deposits amplifies certain periods and damps others.

Bucharest stands on exactly such a sedimentary basin, and the intermediate-depth Vrancea earthquakes produce there a motion rich in long periods, around a second and above. That is precisely the range in which the natural period of ten- to fifteen-storey buildings falls — and it is the reason, put simply, why in 1977 buildings of a particular height suffered disproportionately.

The right way to put it for a class is the one the game teaches anyway: *what matters is not only how hard the ground shakes, but whether its rhythm matches the building.* And the engineering remedy is exactly to change that match — either by stiffening, which shortens the period, or by base isolation, which lengthens it greatly and moves the building out of the dangerous range.

5. Running the lesson

1. **The prediction (3 min).** Before playing: *if you shake harder, do more buildings dance or fewer? And which building do you think needs the slowest rhythm — the tower or the house?* The first question is the activity's deliberate trap.
2. **One projected round (3 min).** Play it yourself, showing *deliberately* first a hard, chaotic shake — which wins no star — and then a slow, regular motion, which lifts the tower. **This contrast is the lesson**, and it is worth showing before any explanation.

Technical note — the three buildings, the star rule and how to hold the box

The sensor. MPU-6500. The game uses **sideways acceleration**, so the box is held **flat** and moved *from side to side* — it is not tilted and not rotated.
How to start. From the **Activities** page, on the **Resonance** card. You choose a box

when it opens.
The three buildings and their frequencies. On screen, each has its name, its natural frequency, its number of storeys and **its natural period, written out**: **The Tower**, 10 storeys, $f_0 = 1.0$ Hz, $T = 1.0$ s; **The Block**, 5 storeys, $f_0 = 2.0$ Hz, $T = 0.5$ s; **The House**, 3 storeys, $f_0 = 3.3$ Hz, $T = 0.3$ s. All three follow the rule $T \approx N/10$. **The period on screen is what the hand must beat** — one second there and back for the tower. Above the buildings, top left, the app also shows the rhythm of your hand — **You are shaking at ≈ ... Hz**, worked out from the last second and a half of motion — to compare directly with f_0. Under the buildings, a panel explains the rule $T \approx N/10$ and $f_0 = 1/T$. When all three have their star, the screen says **Each building danced on its own! You are a seismic engineer!**
The amplitude bar under each building turns red when that building is *dancing*. Below the threshold it is green.
The star rule — read it before you explain the game. A star is won when a building crosses the dance threshold **and, at the same time, has an amplitude at least** 1.6 **times that of the liveliest of the other two**, sustained for **one and two tenths of a second** (on screen: "over a second"). **The rule is also written on screen, under the header**, next to the advice to hold the box flat: *A building wins its star when it dances and moves at least 1.6 times more than either of the other two, for over a second. Shaking hard and wildly moves all three and wins nothing: find each building's own rhythm.* **It is not won by amplitude but by *selectivity***: a hard, disorderly shake excites all three and wins nothing. This is the rule that makes the game a lesson in resonance rather than one in strength.
Resonance builds up over time. A building's amplitude grows progressively, over several seconds of steady rhythm. **It does not appear at once**, so a rhythm held for one second produces nothing — and that is itself a good observation: energy accumulates.
The buttons. Start over clears the stars; × closes the window. There is no export — the outcome of this activity is not a file but the three rhythms found.
WiFi, not the cable. The motion is broad and repeated; the cable tangles and tugs at the box.

3. **The pupils' rounds (8 min).** Taking turns, with the goal stated aloud: *all three stars.* Those waiting have roles — see the adaptations box.

4. **The *harder does not help* test (3 min).** One pupil shakes as hard and as disorderly as they can, for thirty seconds. **The result is no stars**, and the whole class sees why: all three bars rise together and none pulls ahead.

5. **Measuring the periods (full route).** With the metronome or by counting: hold the rhythm at 60 beats per minute and see which building answers; then at 120; then at 200. Fill in the rhythm-and-building table.

6. **The link with real buildings (full route).** The $T \approx N/10$ rule, applied to blocks of flats around town. Which rhythm would be dangerous for each?

Debrief: the score is not the point

Do not ask who got three stars. Ask **"what did you do differently when it worked?"**
The answers pupils give are exactly the content of the lesson: *I slowed down*; *I moved less, but more regularly*; *I counted in my head*; *I kept the same rhythm for longer.* Each of these is a correct description of resonance, in their own words.
The four questions, in order:

- *Did shaking harder help?* — no. And why it did not is the very definition of resonance:

what counts is the match, not the energy.
- *Which building asks for the slowest rhythm, and why that one?* — the tower, because it is the tallest; large mass and long period go together.
- *Why does the dancing not start at once?* — because the amplitude accumulates: each well-timed push adds a little, as on a swing.
- *What does this mean for real buildings in an earthquake?* — a building is in greater danger if the ground's rhythm matches its own. **Not all buildings suffer alike in the same earthquake.**

The score is not the objective. A pupil who won one star but can explain why chaotic shaking does not work has understood the whole lesson. Say so *before* the first round.
And a note on tone. In Romania, talk of buildings falling down is not abstract. Hold the discussion as one about *engineering and preparedness* — what can be done, and what is done — not as one about catastrophe. The *Earthquake!* activity continues in exactly that direction.

The investigation, in one table

The question. What decides which building dances — how hard I shake, or how fast?
The hypothesis. Written **before** playing: if I shake harder, _____ buildings will dance. It is not changed afterwards.
What I change. One thing at a time: first the *strength*, at the same rhythm; then the *rhythm*, at the same strength.
What I observe. Which building has the tallest bar, and whether any star appears.
What I keep the same. The box **flat**, moved sideways only; the same hand amplitude while I vary the rhythm; and **at least five seconds of steady rhythm** before judging the result, because the amplitude accumulates.
How many repeats. Each rhythm twice. If a different building comes up the second time, the rhythm was not steady — which is itself the commonest outcome of a first attempt.
What makes an observation sound — and what makes it an artefact. Sound is an observation made at a steady rhythm, held long enough, with the box flat. An artefact is anything else: **a shake whose rhythm changes as it goes excites each building in turn** and seems to show that "they all dance"; a box tilted instead of moved sideways gives a mixed signal; and a conclusion drawn after one second of rhythm is drawn before the resonance has had time to form.

6. After the activity

6.1 What the observations should look like

What I do	What happens	What it means
Shake hard and chaotically, 30 s	All three bars rise; **no star**	Plenty of energy, no match.
Slow rhythm, about 1 s there and back	The tower grows alone	Resonance at $f_0 = 1$ Hz.
Rhythm twice as fast	The block grows alone	$f_0 = 2$ Hz.
Over three times as fast	The house grows alone	$f_0 = 3.3$ Hz.
Right rhythm, held 1 s	Almost nothing	Resonance needs time to build.
Right rhythm, held 5–10 s	Red bar, then the star	The amplitude grows progressively.
Small amplitude, exact rhythm	**It works**	Amplification at resonance is about tenfold.
Large amplitude, wrong rhythm	It does not work	**The conclusion of the activity.**

6.2 From the period on screen to the rhythm of the hand, step by step

1. **Read the period off the screen.** The tower: $T = 1.0$ s.
2. **Translate it into motion.** One period means *one complete there-and-back*: left, right, back to the middle. So one second for the whole cycle, not for each direction. **This is the commonest confusion**, and it makes the rhythm twice too fast.
3. **Check with the metronome.** 60 beats per minute means 1 Hz. If you move the box fully, there and back, on every beat, you are driving the tower.
4. **Work out the others too.** $T = 0.5$ s $\Rightarrow$ $f = 2$ Hz $\Rightarrow$ 120 beats per minute. $T = 0.3$ s $\Rightarrow$ about 3.3 Hz $\Rightarrow$ about 200 beats.
5. **Apply the $T \approx N/10$ rule** to a real building: an eight-storey block has a period of about 0.8 s, so 1.25 Hz.

6.3 Questions to ask afterwards

1. You shook as hard as you could and won no star. Why did strength not help?
2. The tower has 10 storeys and a period of one second; the house has 3 storeys and a period of three tenths. What connection do you see, and how would you state it as a rule?
3. Why does a building not dance the moment you hit its rhythm? What is happening in the first few seconds?
4. The block you live in has _____ storeys. What period do you estimate, and which shaking rhythm would be most dangerous for it?
5. An engineer wants to move a building out of the dangerous range. They can stiffen it or set it on isolators. What does each solution do to its period?

6.4 Assessment ideas as you go

- **Low stakes:** the pupil ranks the three buildings from the slowest rhythm to the fastest and says what the slow ones have in common.
- **Standard:** the pupil explains why hard shaking wins no stars, using the words "match" or "rhythm", and correctly converts the period on screen into beats per minute.
- **Extension:** the pupil applies the $T \approx N/10$ rule to a real building, estimates its frequency, and explains, using the spectrum of the ground motion, why not all buildings suffer alike in the same earthquake.

Common pitfalls and misconceptions

"I'll shake harder to make it work better." This is the activity's deliberate trap and should be allowed to happen — but it must also be named at the debrief, or it stays a frustration instead of becoming a discovery.
Confusing the period with half of it. A period is a *complete there-and-back*. Someone who moves left once a second is really beating 0.5 Hz and waking a different building from the one they think.
Tilting the box instead of moving it sideways. The game reads sideways acceleration. The box is held flat and *translated*.
Conclusions drawn after one second. Resonance builds over several seconds. Hold the rhythm for at least five.
A rhythm that changes as it goes. It excites the buildings in turn and looks as though "they all dance". A steady rhythm is the condition of the experiment.
Turning it into a league table. Whoever wins three stars fastest has not necessarily understood more; whoever explains why force fails, has.
The USB cable. The motion is broad and repeated; the cable twists and adds motions of its own. Use WiFi.

Safety and classroom management

A wrist or neck strap is recommended. The motion is broad and quick, and the box can slip out of a hand.
A metre of clear space around the player, in every direction. Broad sideways motion strikes classmates and objects on the table.
The player stays put, preferably seated, with the box above the table. No playing standing on a chair, and no walking about the room.
Glasses, tumblers and objects on the table are moved aside before the round.
If you use the pendulums as a physical demonstration: tie the weights securely, use light objects, and move the ruler smoothly rather than jerking it — a heavy pendulum coming loose is the only real hazard of the physical version.
The tone of the discussion. In Romania, the subject of buildings damaged by earthquakes touches family experience. Hold the discussion as one about engineering and preparedness, announce the subject in advance, and leave room for questions.

Adaptations for pupils with special educational needs

Someone else keeps the rhythm. This is the main adaptation and it works very well: one pupil beats the rhythm with a hand or a metronome while another only moves the box, following the beat. **The task splits in two**, and both halves are real — keeping a steady rhythm is precisely the hard skill of this activity.
The metronome instead of counting. For pupils for whom counting in the head consumes all their attention, a metronome at 60, 120 and 200 does the rhythm work, leaving

them free to watch the bars.
Watching the bars instead of playing. A pupil who cannot hold the box takes the role of **bar watcher**: saying aloud which building is growing and when it crosses the threshold. This is the information the whole debrief depends on.
Three real roles, rotated each round. The **rhythm keeper** (hand or metronome); the **bar watcher**; the **scribe**, who records which rhythm woke which building.
The motion on screen. Swaying buildings can be unpleasant for pupils sensitive to motion. Seat them to one side or further back and give them the rhythm keeper's role, which does not require a fixed gaze.

If things do not go to plan

No star lights up, however hard they try. Almost certainly the rhythm is not steady, or is too short. Use the **metronome** at 60 beats per minute and move the box fully there and back on every beat, for ten seconds. If the tower still does not answer, check that the box is being moved *sideways*, flat, rather than tilted.
All three dance at once. The shaking is chaotic. **This is the expected outcome of the *harder does not help* test** — use it as such.
The game stutters. On weak WiFi or with a twisted cable. The game needs frequent data to integrate the oscillators.
One box and thirty pupils. Project the screen, keep the rounds short, and give the three roles to those waiting. The **someone else keeps the rhythm** variant puts two pupils to work on the same round.
The box will not connect. Do not stop the lesson — do the physical version: **three pendulums of different lengths tied to the same ruler**. Move the ruler slowly and only the long one dances; move it quickly and only the short one does. **It is exactly the same phenomenon with no apparatus at all**, and some teachers prefer it. Return to the game another time, once you have checked the box and the network.

7. Further resources

Stable, well-established starting points. Search each site for the subject named — these are top-level addresses, not deep links, so they do not rot.

- **National Institute for Earth Physics (Romania)** — `https://www.infp.ro` — search for the seismic hazard maps and response spectra for Romania; they show why long periods matter in Bucharest.
- **PhET Interactive Simulations** — `https://phet.colorado.edu` — search for the simulations on oscillation and resonance; excellent to show straight after the game.
- **Institute of Physics** — `https://www.iop.org` — search for the material on forced oscillation and resonance, written for teachers.
- **USGS Earthquake Hazards** — `https://www.usgs.gov` — search for building response to earthquakes and site effects; it explains amplification on soft ground.
- **General Inspectorate for Emergency Situations (Romania)** — `https://www.igsu.ro` — for the natural continuation into *Earthquake!* and the preparedness side.
- **STEM Learning** — `https://www.stem.org.uk` — search for resonance and earthquake-resistant design for secondary; many versions of the tower built from straws and modelling clay, which go well alongside this game.

8. Assessment

The exit ticket, identical to the one on the pupil sheet: *why does shaking harder not help?* Collect it in the last two minutes; in this activity it says far more than the number of stars.

For a more formal judgement, use the **shared three-strand rubric** — *measurement practice*, *data interpretation* and *scientific reasoning*. Here the three strands look like this:

- **Measurement practice** — the pupil holds the box flat, moves it sideways, keeps a steady rhythm for at least five seconds and changes one thing at a time.
- **Data interpretation** — the pupil reads the period off the screen, converts it correctly into beats per minute, and matches each rhythm to its building.
- **Scientific reasoning** — the pupil explains why force does not help, links height to period, and applies the idea to a real building.

The rubric itself, with the descriptors for all three levels, is defined once in the *curriculum plan* and is not repeated in the guides.

9. The pupil sheet

The sheet the pupils fill in is a separate document: `docs/en/student_sheets/resonance.pdf`. It runs to two pages — the prediction and the rhythm-and-building table on the first; the *harder does not help* test, the frequency calculation and the conclusion on the second — so it prints double-sided, one sheet per pupil.

Three things to know before printing it. **The sheet explicitly asks for the result of the "I shake as hard as I can" test**, with a row of its own, because *no stars* is a result, and if there is nowhere to write it nobody treats it as one. **The period-to-beats-per-minute column is printed with one row already worked out**, because confusing the period with half of it is the commonest mistake. **The last row of the sheet asks for the number of storeys of the pupil's own block of flats** and the period estimated from the $T \approx N/10$ rule — it is the only place in the whole collection where a pupil applies a formula to their own home. The tables are not repeated here precisely so that the two documents cannot drift apart: there is one printable version, and it is the sheet.